

TABLE 3
COMPARISON OF DIFFERENT IMAGING SENSORS

Model	Company	Type	Chroma	Resolution	Mega-pixels	Max Frame rate	Shutter type	Sensor size	Interface	pixel size (in μm)	Usage
XGS 16000	On Semi	CMOS	Colour, Monochrome	4000 x 4000	16	65	Global shutter	29mm x 29mm	HiSpi	3.2 x 3.2	Machine vision
IMX 487	Sony	CMOS	Colour, Ultraviolet	2856 x 2912	8.13	194	Global shutter	20mm x 16.8mm	SLVS-EC	2.74 x 2.74	Machine vision
GTOF0503	G-pixel	CMOS	Monochrome	640 x 480	-	30	Global shutter	5.3mm x 4mm	MIPI CSI-2	5 x 5	Machine vision
AR0821CS	On Semi	CMOS	Colour	3848 x 2168	8.3	60	Electronic rolling shutter	-	MIPI	2.1 x 2.1	Surveillance, Drone, Machine vision
imx 661	Sony	CMOS	Colour, Black & White	13400 x 9528	127	21.8	Global shutter	75mm x 63.6mm	SLVS-EC	3.45 x 3.45	Machine vision
ISOCeLL HP1	Sam-sung	CMOS	Colour	16384 x 12288	200		Electronic rolling shutter and Global reset	1/1.22"	D-PHY, C-PHY	0.64 x 0.64	Mobile
OVB0B	Omni-vision	CMOS	Colour	16384 x 12288	200	90	Rolling shutter	1/1.28"	mipi	0.61 x 0.61	Mobile
GN5	Sam-sung	CMOS	Colour	8160 x 6144	50	90	Global shutter	1/1.57"	D-PHY, C-PHY	1 x 1	Mobile
Gjoo422	Giga-jot	CMOS	Mono-chrome, Colour	2048 x 2048	4.2	60	Rolling shutter	1/1.28"	28 LVDS	2.2 x 2.2	Scientific, astronomy

readers, and surveillance systems.

Sony IMX 661. Sony launched the 127MP IMX 661 in the first quarter of 2021. When launched, this sensor had the highest pixel count for any industrial CMOS sensor equipped with a global shutter. This

image sensor features an astounding resolution of 13400×9528 pixels with a maximum frame rate 21.8fps. The imaging sensor is quite a lot bigger than any other sensors on the list with its diagonal size measuring 56.73mm. It features multiple exposures, trigger synchronisation, and Pregius Global Shutter technology, which allows the camera to take high-resolution images with very low distortion.

Samsung ISOCeLL HP1.

The ISOCeLL HP1, launched in the third quarter of 2021, is the industry's earliest

200MP sensor specially developed for handheld devices, such as smartphones. This sensor features a small pixel of $0.64\mu\text{m}$ and employs Chameleon Cell pixel-binning technology. Depending on the ambient lighting, Chameleon pixel-binning can capture full resolution pics or employ a 2×2 pixel binning with an output of 50MP picture, or even 4×4 binning to provide a 12MP image in extremely low-light conditions. The sensor can capture 8k videos at 30fps and can deliver data at rate up to 4.5Gbps per lane using D-PHY interface.

Samsung GN5.

Samsung launched the GN5 imaging sensor alongside the HP1 in the first week of September 2021. It is a 50MP sensor with extremely fast autofocus capabilities. Samsung claims that the GN5 is the industry's first $1.0\mu\text{m}$ sensor with a Dual Pixel Pro technology, which can place dual photodiodes and recognise pattern changes in all directions to enable an all directional autofocus. It also has Samsung's proprietary pixel technology. Sam-

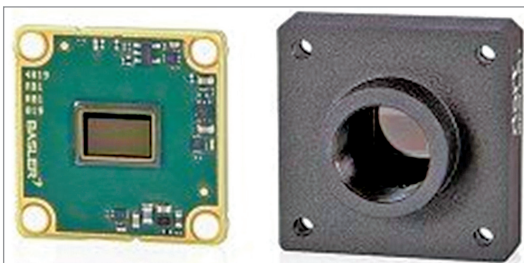


Fig. 12: Basler daA3840-3 with AR0821CS (Source: Link)

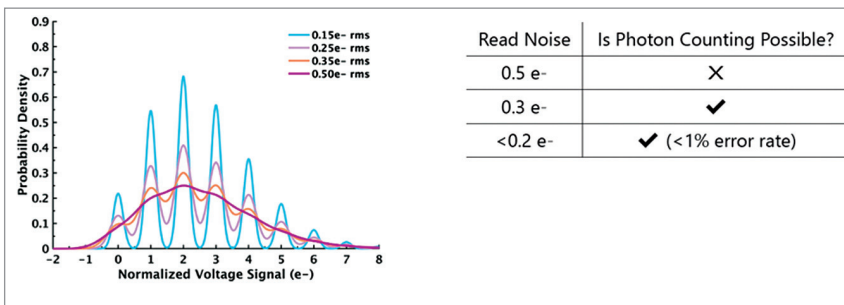


Fig. 13: Photon counting capability at different noise ratio (Source: Gigajot)